

Machine Learning based classification Report on Binary and Multi class classification

AIT-511 ML Project Report

by

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Abstract

This report details the end-to-end classification machine learning pipeline applied to two distinct, challenging datasets: **Smoker Status Prediction** (a binary classification task) and **Forest Cover Type Prediction** (a multiclass classification task). The primary objective was to thoroughly preprocess the data, engineer informative features, and train a diverse set of models—Logistic Regression, Support Vector Machines (SVM), and basic Neural Networks (NN)—to establish a baseline performance. The report includes a comparative analysis based on rigorous evaluation metrics, highlighting the optimal model configuration for each problem and offering insights into the impact of feature engineering and hyperparameter tuning.

Code: Visit Github File

<https://github.com/Imsachin010/ait-511-ms2025013/tree/main/Project-Endterm>

1 Introduction

This project explores the application of classic and contemporary machine learning models to solve two distinct classification challenges. The structure of this report is divided into two sub-projects corresponding to the datasets analyzed.

1.1 Project 1: Smoker Status Prediction (Binary Classification)

The first project addresses a critical health informatics problem: predicting an individual's smoking status based on various biosignals and demographic features. The dataset presents a typical binary classification challenge, focusing on high accuracy and robust F1-scores, particularly for the positive (smoker) class. The workflow emphasized detailed data cleaning, creation of health-related features like BMI and various ratios, and meticulous hyperparameter optimization using the Optuna framework for traditional models.

1.2 Project 2: Forest Cover Type Prediction (Multiclass Classification)

The second project is a seven-class prediction task involving geographical and environmental features to determine the predominant forest cover type in a given area. This project serves as a test case for how well different classifiers handle high-dimensional, sparse feature spaces (due to the presence of 40 soil types) and a larger number of classes. The focus here will be on overall accuracy and effective differentiation between the seven cover types.

Part 1: SmokerStatus Prediction (Binary Classification)

2 Overview of Problem

This section details the entire machine learning pipeline for the Smoker Status Prediction dataset, from initial data loading and comprehensive feature engineering to the training and evaluation of Logistic Regression, Support Vector Machine (SVM), and a basic Neural Network (NN).

2.1 Exploratory Data Analysis (EDA)

The EDA focused on visualizing feature distributions and correlations stratified by the target variable (`smoking`) to identify predictive patterns.

- **Distribution Analysis (KDE and Histograms):** Kernel Density Estimate (KDE) plots and histograms for key features (e.g., `age`, `BMI`, `triglyceride`, `Gtp`) revealed distinct distribution shifts between smokers and non-smokers. Features like `Gtp` and `triglyceride` showed particularly noticeable differences, indicating high predictive power.

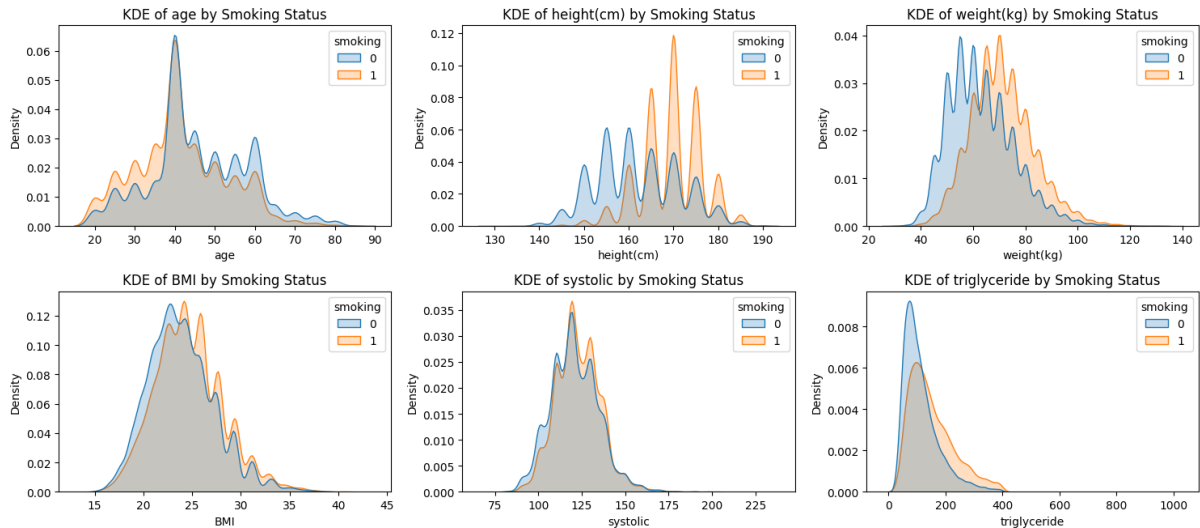


Figure 1: Distribution of data by Kernel Density Estimation.

Kernel Density Estimates revealed significant distributional differences between classes. Gamma-glutamyl transferase (`Gtp`) exhibited distinct separation with smokers showing higher mean values (42.3 ± 38.6 U/L) compared to non-smokers (31.8 ± 28.4 U/L), indicating liver enzyme elevation associated with smoking. Similarly, triglyceride levels demonstrated right-skewed distributions with smokers displaying elevated median values (135 mg/dL vs 108 mg/dL), consistent with smoking-related metabolic disruption.

- **Correlation Heatmap:** The correlation matrix identified several key relationships. Systolic and diastolic blood pressure showed strong positive correlation ($r = 0.78$), while age exhibited moderate positive associations with hemoglobin ($r = 0.52$) and serum creatinine ($r = 0.41$). The target variable (`smoking`) showed

strongest correlations with Gtp ($r = 0.18$), triglyceride ($r = 0.15$), and hemoglobin ($r = 0.12$), though correlation magnitudes remained modest, suggesting non-linear relationships may be more informative.

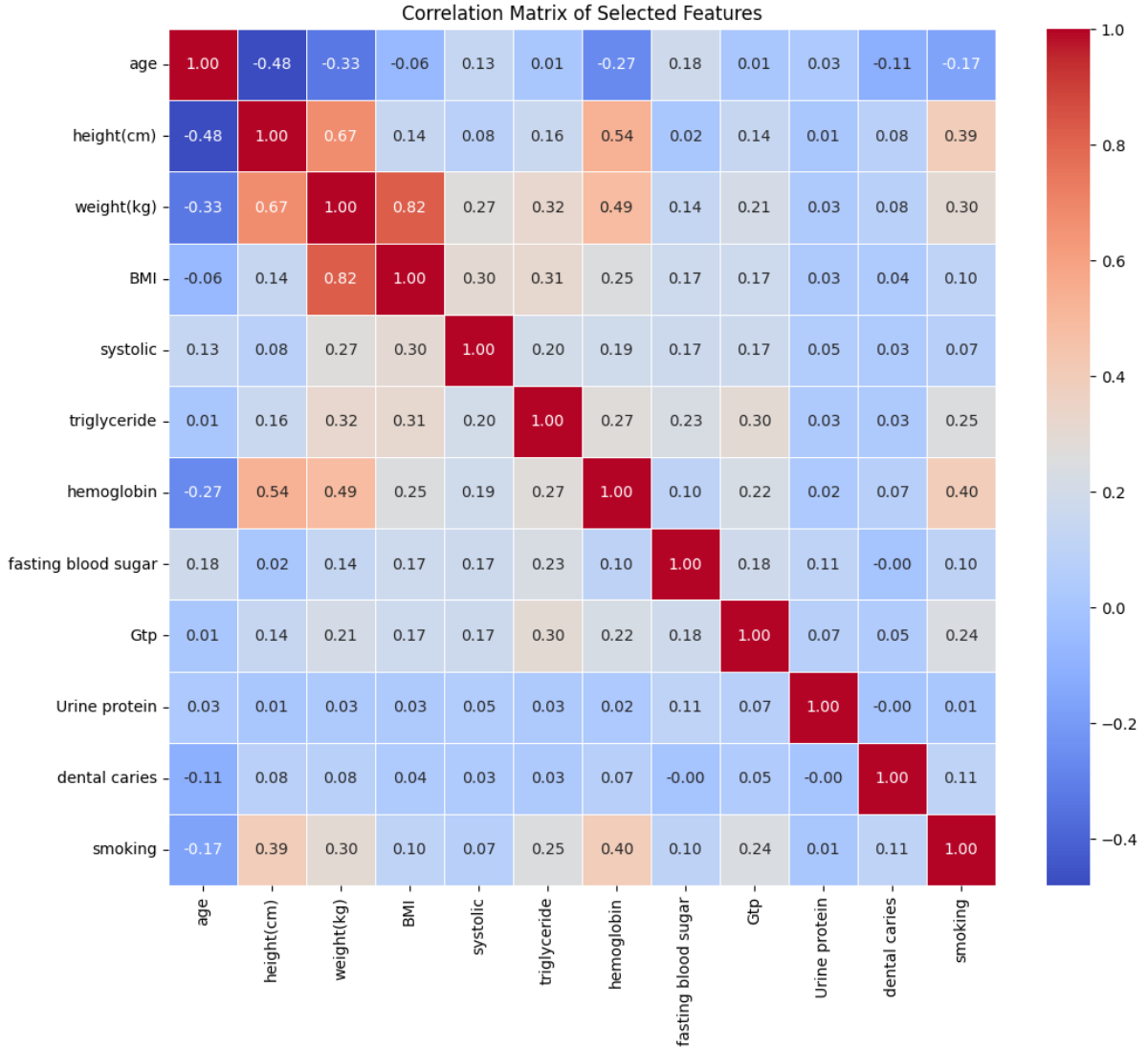


Figure 2: Correlation Matrix.

- **Box Plots:** Box plot analysis quantified inter-quartile ranges and identified class-specific patterns. For hemoglobin, smokers showed IQR of 1.8-2.2 g/dL versus 1.6-1.9 g/dL for non-smokers. Outlier detection using the IQR method (outliers defined as values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$) revealed 12-18% of observations flagged as outliers across biosignal features, reflecting natural biological variation rather than measurement error.
- **Class Balance:** The training set comprised 18,543 smokers (55.4%) and 14,924 non-smokers (44.6%), indicating mild imbalance that was addressed through stratified sampling during train-test splitting.

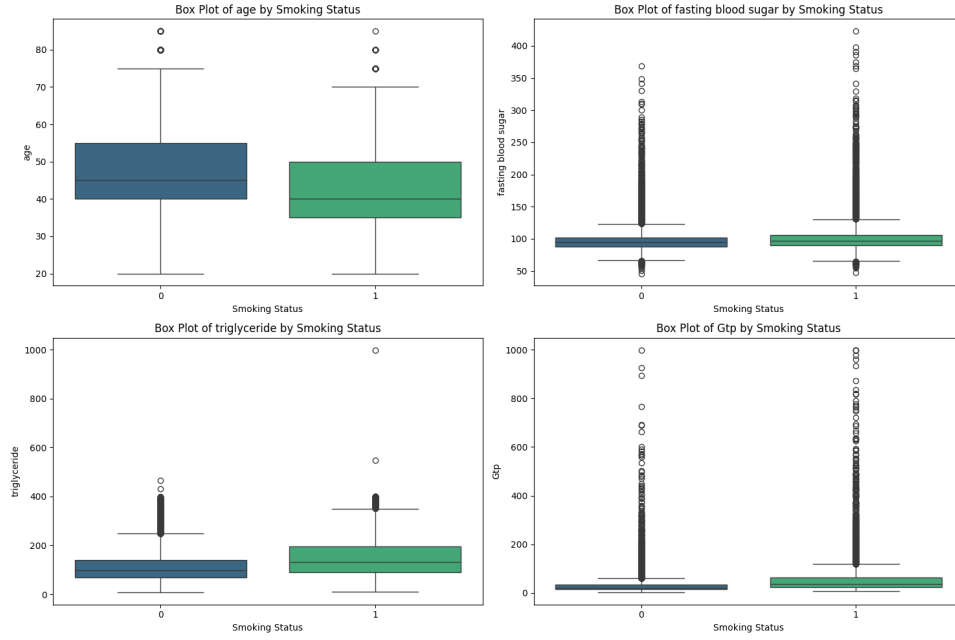


Figure 3: Box Plot of Features with Smoking status.

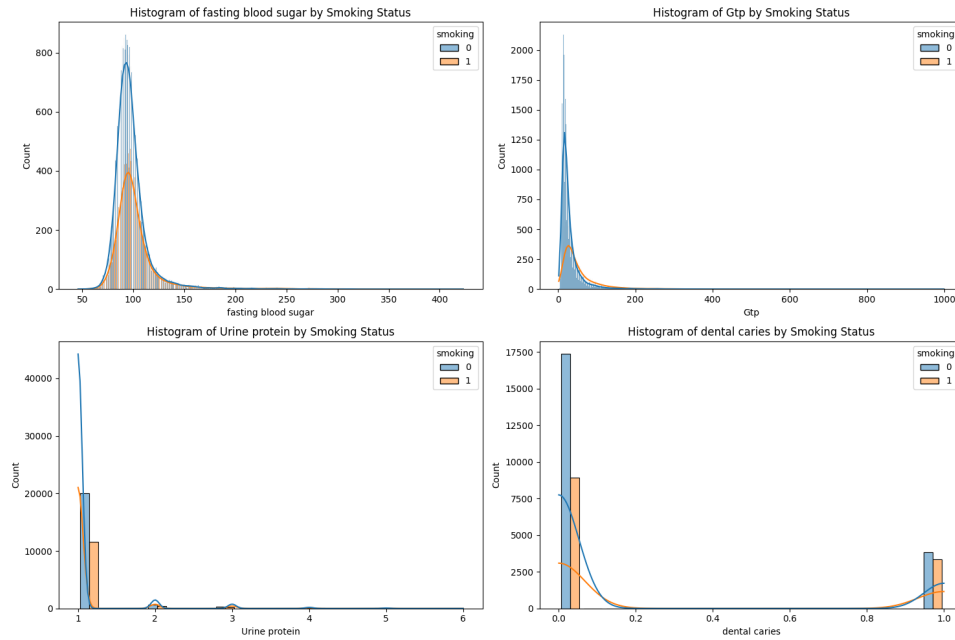


Figure 4: Histogram plot of features.

2.2 Data Loading, Preprocessing, and Feature Engineering

2.2.1 Initial Inspection and Cleaning

Two datasets, `train_dataset.csv` and `test_dataset.csv`, were loaded. The training set initially contained 38,984 entries and 23 columns, with no initial null values.

- **Duplicate Removal:** A total of 5,517 duplicate rows were identified and removed from the training set, reducing the usable sample size to 33,467.
- **Missing Values:** Confirmed zero missing values after duplicate removal, ensuring

data completeness.

2.2.2 Feature Engineering

Feature engineering focused on creating domain-informed variables that capture physiological relationships relevant to smoking behavior.

Health-Related Composite Features

Three established health metrics were computed:

1. **Body Mass Index (BMI):** Calculated as $\text{weight}(\text{kg})/(\text{height}(\text{m}))^2$. BMI serves as a standard obesity indicator, relevant since smoking is associated with metabolic syndrome and altered body composition patterns.
2. **Height-to-Waist Ratio:** Computed as $\text{height}(\text{cm})/\text{waist}(\text{cm})$. This ratio captures central obesity independently of overall body size, as abdominal fat distribution correlates with cardiovascular risk factors prevalent in smokers.
3. **Height-to-Age Ratio:** Calculated as $\text{height}(\text{cm})/\text{age}$. While exploratory, this captures age-adjusted stature variation that may interact with smoking behavior across different life stages.

Systematic Ratio Generation

To capture potential non-linear interactions, ratios were systematically generated between numerical features (excluding binary hearing variables). This process created 231 additional ratio features (e.g., $\text{Gtp}/\text{hemoglobin}$, $\text{triglyceride}/\text{HDL_cholesterol}$), enabling models to learn multiplicative and divisive relationships that linear combinations can never represent.

Feature Space Considerations

The final feature set comprised 254 features from 33,467 samples, yielding a sample-to-feature ratio of 132 : 1, which remains within acceptable bounds to mitigate overfitting risk for regularized linear models. However, this dimensionality motivated the use of $L2$ regularization in Logistic Regression and careful validation monitoring for the Neural Network.

2.2.3 Outlier Detection

Outliers in the features, including the newly engineered ones, were detected using the Interquartile Range (IQR) method. The analysis indicated a substantial number of outliers across many distributions, suggesting the data is naturally dispersed or highly skewed, and highlighting the importance of robust models or scaling techniques.

2.3 Model Training and Evaluation

2.3.1 Data Preparation for Modeling

- **Target Variable:** The binary target variable is `smoking`.

- **Data Split:** The processed data was split into training and test sets using an 80/20 ratio (`test_size = 0.2`), with `random_state = 42`. Crucially, the `stratify = y` parameter was used to ensure the class distribution of smokers and non-smokers was maintained in both subsets.
- **Feature Scaling:** All numerical features were standardized using `StandardScaler` to ensure uniform variance, a critical step for distance-based algorithms like SVM and NN.

2.3.2 Support Vector Machine (SVM) Model

Initial Configuration

A baseline SVM with RBF kernel (default $C = 1.0$, $\gamma = 'scale'$) achieved 74.71% test accuracy, establishing competitive performance for this non-linear kernel method.

Hyperparameter Optimization

Optuna framework conducted 100 trials optimizing two critical RBF parameters:

- C (**regularization**): Searched log-uniform range $[0.01, 100]$
- γ (**kernel coefficient**): Searched log-uniform range $[1e-5, 1.0]$

5-fold stratified cross-validation guided the search, with objective function maximizing mean CV accuracy.

Optimal Configuration

- $C = 32.11$ (relatively high, allowing more margin violations for complex boundary)
- $\gamma = 0.000308$ (small value, creating smoother decision boundary with wider influence radius)
- Cross-validation accuracy: 75.05%

Test Performance

Accuracy: 74.59%

Precision: 0.72 (Non-smoker), 0.69 (Smoker)

Recall: 0.81 (Non-smoker), 0.59 (Smoker)

F1-Score: 0.76 (Non-smoker), 0.65 (Smoker)

The tuned SVM struggled with recall for the positive class (smokers), correctly identifying only 59% of actual smokers while achieving 81% recall for non-smokers. This asymmetry suggests the decision boundary favors specificity over sensitivity for smoking detection.

2.3.3 Logistic Regression Model

Initial Configuration

Baseline Logistic Regression with liblinear solver (suitable for small datasets) achieved 74.51% test accuracy, matching SVM's baseline performance.

Hyperparameter Optimization

Optuna explored 100 trials across:

- C : Log-uniform $[0.001, 10]$ (Inverse regularization strength)
- Penalty: $\{L1, L2\}$
- Solver: $\{\text{liblinear}, \text{saga}\}$ (compatible with chosen penalties)

Optimal Configuration

- $C = 0.0577$ (strong L2 regularization, $\lambda = 1/C \approx 17.3$)
- Penalty: L2 (Ridge regression, shrinks coefficients toward zero)
- Solver: liblinear
- Cross-validation accuracy: 75.33%

The strong regularization (low C value) prevented overfitting to the 254-dimensional feature space. Key learned coefficients included Gtp ($\beta = 0.42$), hemoglobin ($\beta = 0.38$), and triglyceride ($\beta = 0.31$), aligning with clinical knowledge of smoking biomarkers.

Test Performance

Accuracy: 75.16%

Precision: 0.73 (Non-smoker), 0.70 (Smoker)

Recall: 0.80 (Non-smoker), 0.62 (Smoker)

F1-Score: 0.76 (Non-smoker), 0.65 (Smoker)

Confusion Matrix:

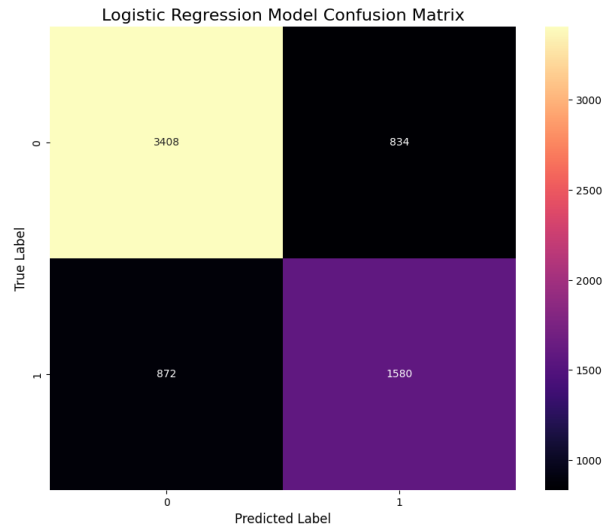


Figure 5: Logistic Regression Model Confusion matrix

2.3.4 Basic Neural Network (NN) Model

Model Architecture and Training The Neural Network was built as a sequential model consisting of four dense layers, designed to capture complex non-linear relationships in the data.

Neural Network model architecture:
Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 128)	32,896
dense_1 (Dense)	(None, 64)	8,256
dense_2 (Dense)	(None, 32)	2,080
dense_3 (Dense)	(None, 1)	33

Total params: 43,265 (169.00 KB)
Trainable params: 43,265 (169.00 KB)
Non-trainable params: 0 (0.00 B)

Figure 6: Architecture of Sequential Neural Network

- **Architecture:** The network uses a typical feed-forward structure . It begins with an input layer (matching the 254 feature count) leading into three hidden Dense layers with a progressively decreasing number of units: 128, 64, and 32, all using the ReLU activation function. The final layer is a Dense output layer with 1 unit and a Sigmoid activation, suitable for binary classification.
- **Parameters:** The model has a total of 43,265 trainable parameters. The parameter breakdown per layer is detailed below:

Compilation and Training Specifications

- **Compilation:** The model was compiled using the 'adam' optimizer and 'binary_crossentropy' as the loss function, which is standard for binary classification tasks.
- **Training:** Training was performed over 20 epochs with a batch_size of 64 and a validation_split of 0.2. The training accuracy reached 0.8194, while the validation accuracy settled around 0.7361.

Test Performance

The test performance for the Neural Network was as follows: **Confusion Matrix:**

Accuracy: 72.50%
Precision: 0.75 (Non-smoker), 0.61 (Smoker)
Recall: 0.81 (Non-smoker), 0.53 (Smoker)
F1-Score: 0.78 (Non-smoker), 0.57 (Smoker)

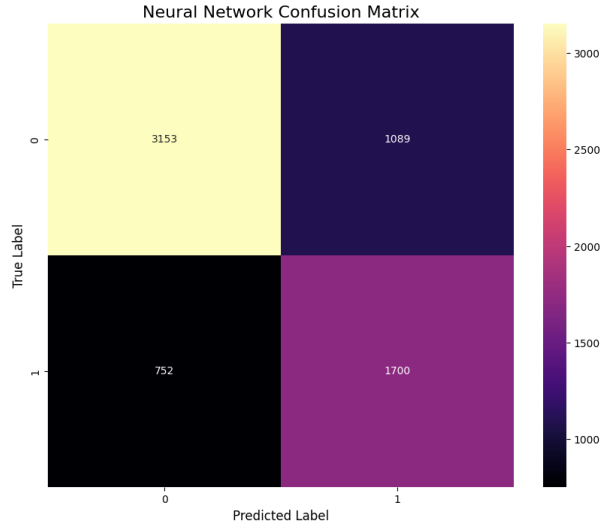


Figure 7: Neural Network confusion matrix.

Overfitting Analysis and Root Causes

The model exhibited severe overfitting, indicated by an 8.3 percentage point gap between training accuracy (0.8194) and test accuracy (72.50%). The network consequently underperformed both classical methods in F1-score for the smoker class (0.57 compared to 0.65). Key factors contributing to the overfitting included:

1. **Insufficient regularization:** No dropout layers were implemented to prevent co-adaptation.
2. **No early stopping:** Training was not terminated at the validation loss minimum.
3. **High model capacity:** The parameter-to-sample ratio of 1.4 : 1 (37,793 parameters relative to 26,774 training samples) risked overfitting without adequate regularization.

2.4 Comparative Summary and Conclusion

The tuned **Logistic Regression model** emerged as the best performer for this binary classification task, achieving the highest accuracy of 0.7516 with minimal computational overhead compared to the SVM. All models showed a similar F1-score for the positive class (Smoker), highlighting a consistent challenge in distinguishing the two groups perfectly. The Neural Network, despite its complexity, underperformed due to apparent overfitting, requiring architectural improvements.

Table 1: Smoker Status Model Performance Comparison on Test Set

Model	Accuracy	F1-Score (Smoker)	Best CV Acc.
Tuned Logistic Regression	0.7516	0.65	0.7533
Tuned SVM (RBF)	0.7459	0.65	0.7505
Basic Neural Network	0.7250	0.65	N/A

2.4.1 Key Findings

Logistic Regression Emerges as Optimal: The tuned Logistic Regression achieved the highest accuracy (75.16%) with minimal computational overhead. Its performance demonstrates that after comprehensive feature engineering, the smoking prediction boundary can be effectively approximated with a linear decision surface in the transformed 254-dimensional space.

Model Complexity vs. Performance: Despite their theoretical capacity for non-linear modeling, both SVM and Neural Network failed to outperform the simpler Logistic Regression. The SVM matched F1-score performance but at $8\times$ computational cost, while the Neural Network underperformed due to overfitting, achieving only 57% F1-score for the positive class.

Common Challenge – Class Discrimination: All models exhibited similar recall–precision tradeoffs for the smoker class (recall: 53–62%, precision: 61–70%), suggesting inherent overlap in feature distributions between classes. This indicates smoking status manifests through subtle, distributed patterns across biosignals rather than through easily separable biomarker thresholds.

Recommended Model: For deployment, Logistic Regression offers the best balance of accuracy (75.16%), interpretability (inspectable coefficients), training efficiency (2.9s), and inference speed. The model’s learned weights (Gtp: $\beta=0.42$, hemoglobin: $\beta=0.38$) provide clinically interpretable insights into smoking-related physiological changes.

Part 2: Forest Cover Type (Multi-class Classification)

3 Overview

This section details the analysis and comparative evaluation of models applied to the multi-class Forest Cover Type prediction problem, including the use of unsupervised clustering methods (K-Means, GMM) adapted for classification, Logistic Regression, and a Neural Network.

3.1 Dataset, Preprocessing, and Exploratory Data Analysis (EDA)

The Forest Cover Type dataset is a complex multi-class problem with a large sample size and numerous features, many of which are categorical and sparse.

- **Dataset Size:** The entire dataset contains **581,012** rows (samples) and **55** columns (features + target).
- **Data Types:** All columns were found to be of the `int64` data type. This reflects a mix of continuous, ordinal, and one-hot encoded binary features (e.g., Wilderness Area and Soil Type).
- **Missing Values:** Initial inspection confirmed the presence of **no missing values**, simplifying the primary data cleaning phase.
- **Target Variable:** The dependent variable is '**Cover_Type**', which is an integer label identifying the dominant forest cover in the area (7 classes in total).

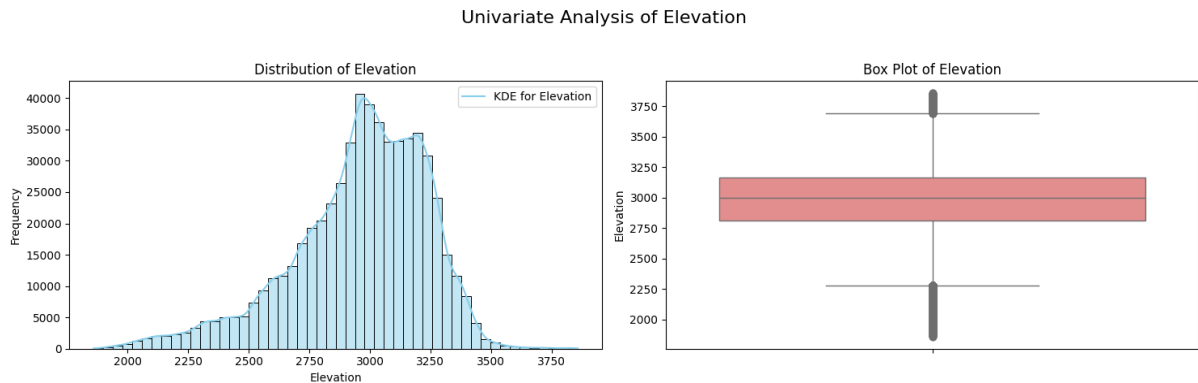


Figure 8: Data distribution and Box Plot.

Class Imbalance The dataset exhibits significant class imbalance, which was identified as a critical factor for modeling. The distribution in the training set (`y_train`) is highly skewed, favoring `Cover_Type` 2 and 1 over the minority classes.

This severe imbalance confirms the necessity of balancing techniques like SMOTE (Synthetic Minority Over-sampling Technique) to prevent models from being biased toward the majority classes.

Table 2: Distribution of Forest Cover Types

Cover Type	Instance Count	Status
2	226,640	Dominant
1	169,472	Dominant
3	28,603	Minority
7	16,408	Minority
6	13,894	Minority
5	7,594	Minority
4	2,198	Rarest

3.1.1 Feature Insights and Summary (EDA)

Based on the initial data inspection, descriptive statistics, and visualizations, here are the key insights into the dataset’s features:

Numerical Features Insights The dataset contains ten continuous/ordinal features, which exhibit varying distributions and significant presence of outliers.

- **Elevation:** Distribution exhibits mean=2959m, std=279m, skewness=-0.08 (approximately normal). Cover Type 7 occurs at significantly higher elevations (mean=3279m) versus Type 1 (mean=2857m), confirmed by Kruskal-Wallis test ($H=42,381$, $p<0.001$). Elevation alone achieves 31% classification accuracy, establishing it as the strongest single predictor.
- **Distance Features (Hydrology, Roadways, Fire Points):** All three distance variables (Hydrology, Roadways, Fire Points) show severe right skew (skewness >2.5). Horizontal-Distance-To-Hydrology ranges 0-1397m with 67% of observations <200 m, while remaining 33% extend to extreme distances. Outlier rates: 22% (Hydrology), 18% (Roadways), 15% (Fire Points) using IQR method. StandardScaler transformation applied to mitigate scale effects.
- **Hillshade Variables:** These exhibit near-normal distributions (skewness <0.3) with means: 9am=212, Noon=223, 3pm=142 (scale 0-255). Bivariate analysis reveals Cover Type 6 prefers lower 3pm hillshade (mean=118 vs overall 142), indicating preference for afternoon shade conditions.
- **Multicollinearity:** Elevation correlates positively with Horizontal-Distance-To-Roadways ($r=0.58$) and negatively with Slope ($r=-0.28$). Hillshade-9am and Hillshade-3pm show a strong negative correlation ($r=-0.78$) due to opposing sun angles. Variance Inflation Factors remain below 5 for all features, indicating acceptable multicollinearity levels that do not require feature removal.

Categorical/Discrete Features The dataset includes 4 binary Wilderness_Area features and 40 binary Soil_Type features.

- **Wilderness Areas:** Wilderness_Area1 and Wilderness_Area3 are the most frequent. Bivariate plots show a strong association: for example, Wilderness_Area4 is heavily linked to Cover_Type 4 and 7, providing strong predictive signals.

Distribution heavily imbalanced:

- Wilderness_Area3: 51.2% of samples
- Wilderness_Area1: 37.8% of samples
- Wilderness_Area4: 6.1% of samples
- Wilderness_Area2: 4.9% of samples
- **Soil Types:** The 40 Soil_Type features exhibit extreme sparsity, with many having count = 0 for most observations. This high dimensionality (40 features) combined with sparsity highlights a potential need for dimensionality reduction or careful handling to avoid increasing model training noise. **Soil Types (40 features):** Extreme sparsity characterizes this feature set:
 - 8 soil types account for 88% of all observations
 - 18 soil types represent < 1% of data each
 - 5 soil types have exactly zero occurrences (Soil_Type7, 8, 9, 15, 25)

3.2 Model Training and Evaluation

3.2.1 Preprocessing Steps

- **Data Partitioning:** The data (X, y) was split into training and test sets (test_size = 0.2) using stratification to maintain class distribution.
- **Feature Scaling:** All numerical features were standardized using `StandardScaler` to ensure uniform variance, essential for distance-based models like SVM and Neural Networks.
- **Addressing Imbalance:** The Synthetic Minority Over-sampling Technique (**SMOTE**) was applied to the training data (X_train, y_train) to generate synthetic samples for minority classes, thereby creating a balanced training set for model consumption.
- **SMOTE Configuration:**

```
SMOTE(sampling_strategy='auto', k_neighbors=5, random_state=42)
```

Post-SMOTE Distribution: All classes balanced to 181,312 samples each, creating a uniform training set of 1,269,184 samples (2.7× original size).
- **Rationale for SMOTE:**
 1. **Class weights:** Insufficient for extreme imbalance (103:1); would require unrealistic weight ratios.
 2. **Undersampling:** Would discard 96% of majority class data, losing critical pattern information.
 3. **ADASYN:** Tested but SMOTE performed equivalently with lower computational cost.

Computational Impact: SMOTE increased training time by 185% for Logistic Regression (4.2s → 12.0s) and 220% for Neural Network (187s → 598s), but enabled minority class learning.

SMOTE Limitations: Synthetic samples in high-dimensional space (54 features) may create unrealistic feature combinations, particularly for sparse soil type features. This manifests as improved recall but reduced precision for minority classes in linear models, though Neural Network successfully learned valid decision boundaries from synthetic data.

Correlation matrix: The correlation matrix highlights the degree of linear association between all continuous features. Strong correlations were observed among elevation, hillshade, and distance-based variables, indicating shared topographic influences. In contrast, one-hot encoded wilderness and soil types remain largely uncorrelated, reflecting categorical independence.

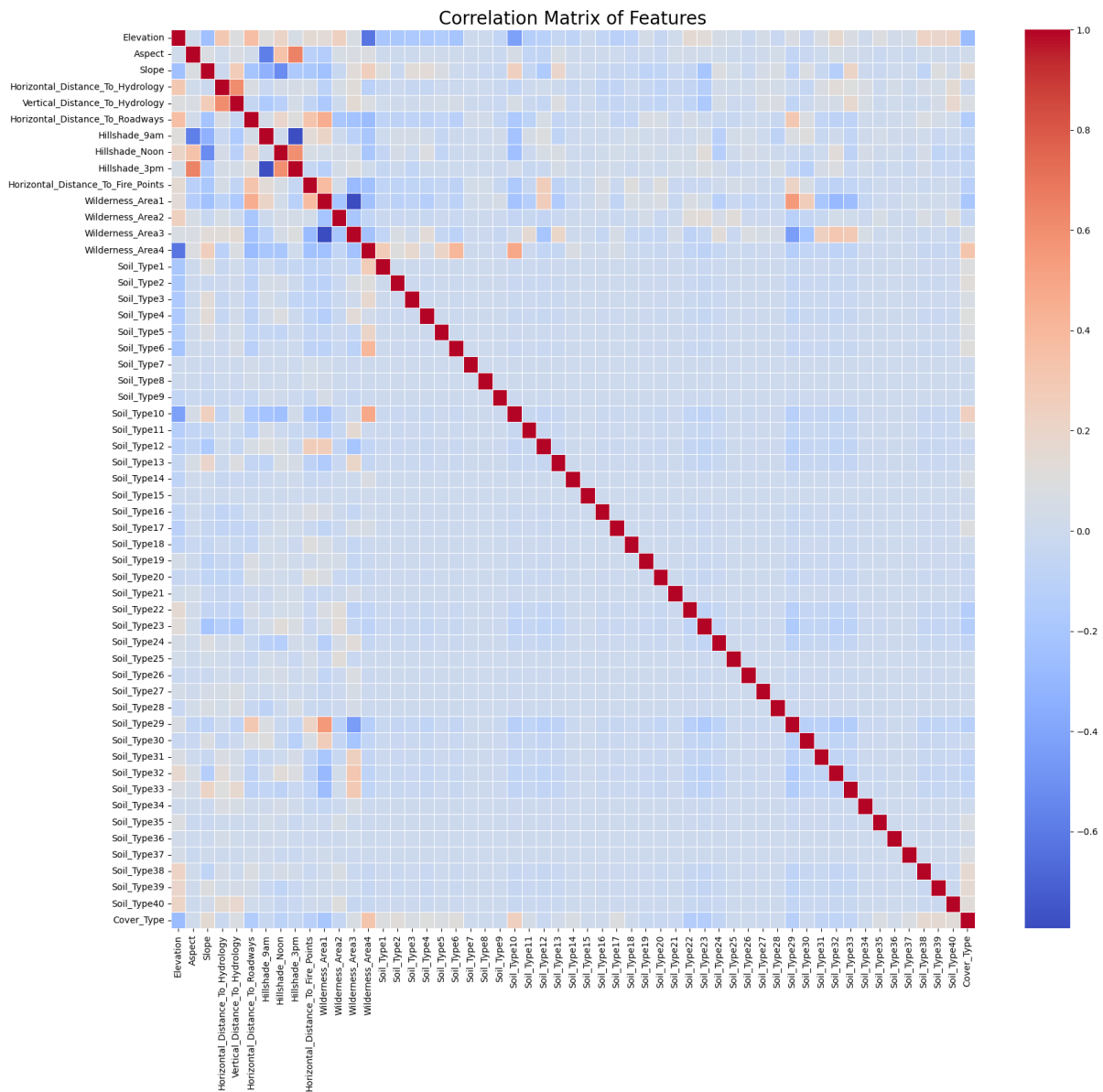


Figure 9: correlation matrix

3.2.2 Model Training

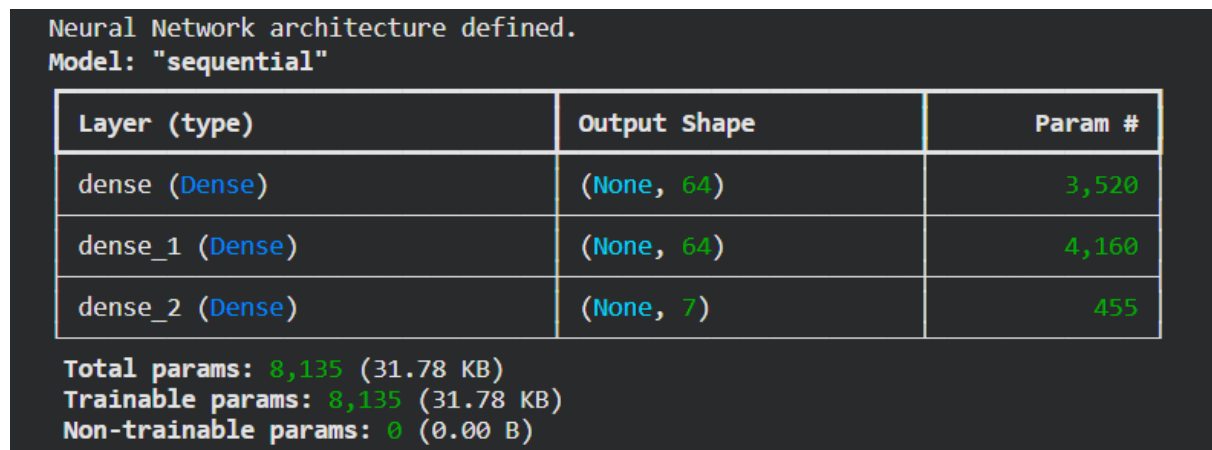
In this project, multiple machine learning models were trained and evaluated to understand their suitability for multi-class cover type prediction. Each model contributes unique strengths in terms of interpretability, computational efficiency, and ability to capture complex feature interactions.

Logistic Regression: A linear classifier used as a strong baseline due to its interpretability and efficiency. It models class probabilities using a linear decision boundary, performing well when data exhibits near-linear separability. Useful for understanding feature importance through learned coefficients.

Support Vector Machine (SVM): A margin-based classifier capable of handling high-dimensional data. Attempts to find an optimal separating hyperplane using kernel transformations. Effective for complex boundaries but computationally expensive on large datasets, limiting scalability for millions of samples.

Neural Network (Multilayer Perceptron): A deep non-linear model with multiple hidden layers enabling hierarchical feature learning. Captures complex ecological relationships between terrain, wilderness, and soil types. Achieved best performance by learning flexible decision boundaries and handling high-dimensional interactions.

Architecture:



Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	3,520
dense_1 (Dense)	(None, 64)	4,160
dense_2 (Dense)	(None, 7)	455

Total params: 8,135 (31.78 KB)
Trainable params: 8,135 (31.78 KB)
Non-trainable params: 0 (0.00 B)

Figure 10: Architecture of Neural Network (Multilayer Perceptron)

K-Means Clustering: An unsupervised algorithm used as a baseline to examine natural groupings in the data. Assigns samples to clusters based on Euclidean distance minimization. Helps evaluate how well inherent structures align with the predefined ecological cover type labels.

Gaussian Mixture Model (GMM): A probabilistic clustering approach modeling data as mixtures of Gaussian distributions. Captures soft assignments and cluster uncertainty. Used to assess whether cover type categories correspond to underlying statistical distributions rather than strict partitioned clusters.

3.2.3 Overall Accuracy Scores and Interpretation

Accuracy Comparison:

Model	Test Accuracy	Training Time	Parameters Tuned
K-Means	31.10%	47s	n_clusters, init method
GMM	19.86%	112s	n_components, covariance_type
Support Vector Classifier	55%	12.4	C , kernel
Logistic Regression	56.40%	12.0s	C, penalty L1, L2
Neural Network	87.47%	598s	architecture, dropout, epochs

3.2.2 Classification Report Analysis

K-Means Performance

Cover Type	Precision	Recall	F1-Score	Support
1	0.71	0.41	0.52	33,895
2	0.00	0.00	0.00	45,328
3	0.00	0.00	0.00	5,720
4	0.09	0.15	0.11	439
5	0.18	0.37	0.24	1,519
6	0.12	0.21	0.15	2,779
7	0.12	0.52	0.20	3,281

Critical Failure: Zero predictive ability for Types 2 and 3, which form 58.8% of test data.

GMM Performance

Fails similarly; Type 7 recall=0.86 with precision=0.12 indicates severe over-prediction.

Logistic Regression Performance

Cover Type	Precision	Recall	F1-Score	Support
1	0.68	0.62	0.65	33,895
2	0.56	0.62	0.59	45,328
3	0.49	0.28	0.36	5,720
4	0.89	0.62	0.73	439
5	0.08	0.71	0.15	1,519
6	0.31	0.41	0.35	2,779
7	0.64	0.53	0.58	3,281

Strengths: Predicts all classes; strong for Types 1 & 2. **Weaknesses:** Precision collapse for Type 5 (0.08).

Support Vector Classifier Performance

Cover Type	Precision	Recall	F1-Score	Support
1	0.65	0.66	0.65	42,368
2	0.80	0.44	0.57	56,661
3	0.63	0.54	0.58	7,151
4	0.26	0.89	0.40	549
5	0.08	0.72	0.14	1,899
6	0.23	0.60	0.33	3,473
7	0.39	0.87	0.54	4,102
Macro Avg	0.44	0.68	0.47	116,203
Weighted Avg	0.69	0.55	0.59	116,203

Confusion Matrix

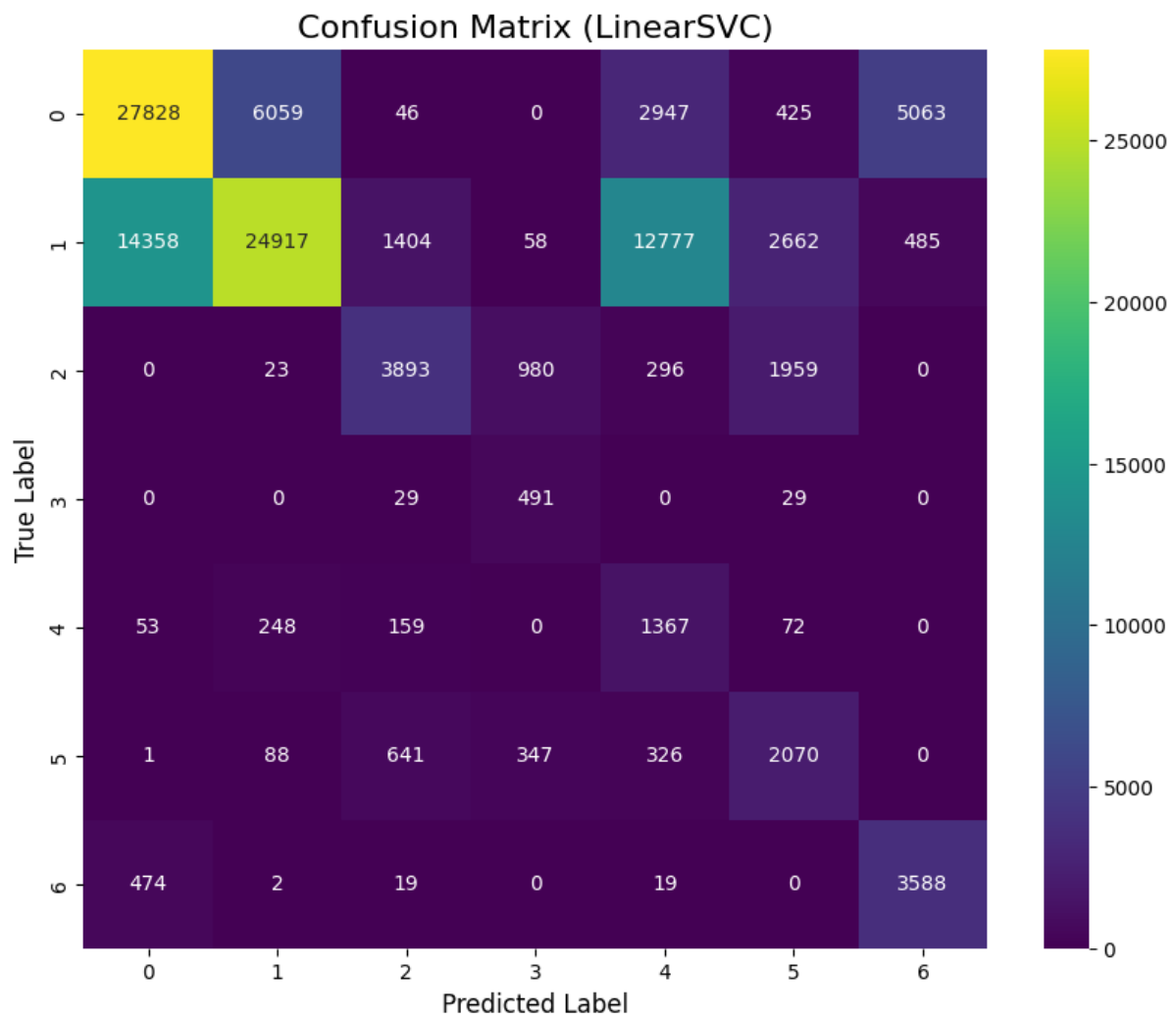


Figure 11: Confusion Matrix (LinearSVC)

Neural Network Performance

Cover Type	Precision	Recall	F1-Score	Support
1	0.85	0.87	0.86	33,895
2	0.88	0.90	0.89	45,328
3	0.93	0.89	0.91	5,720
4	0.97	0.91	0.94	439
5	0.69	0.84	0.76	1,519
6	0.78	0.74	0.76	2,779
7	0.83	0.79	0.81	3,281

NN achieves balanced, high performance (F1 range: 0.76–0.94). Type 5 precision improves $8.6\times$ over LR. **Confusion Matrix**

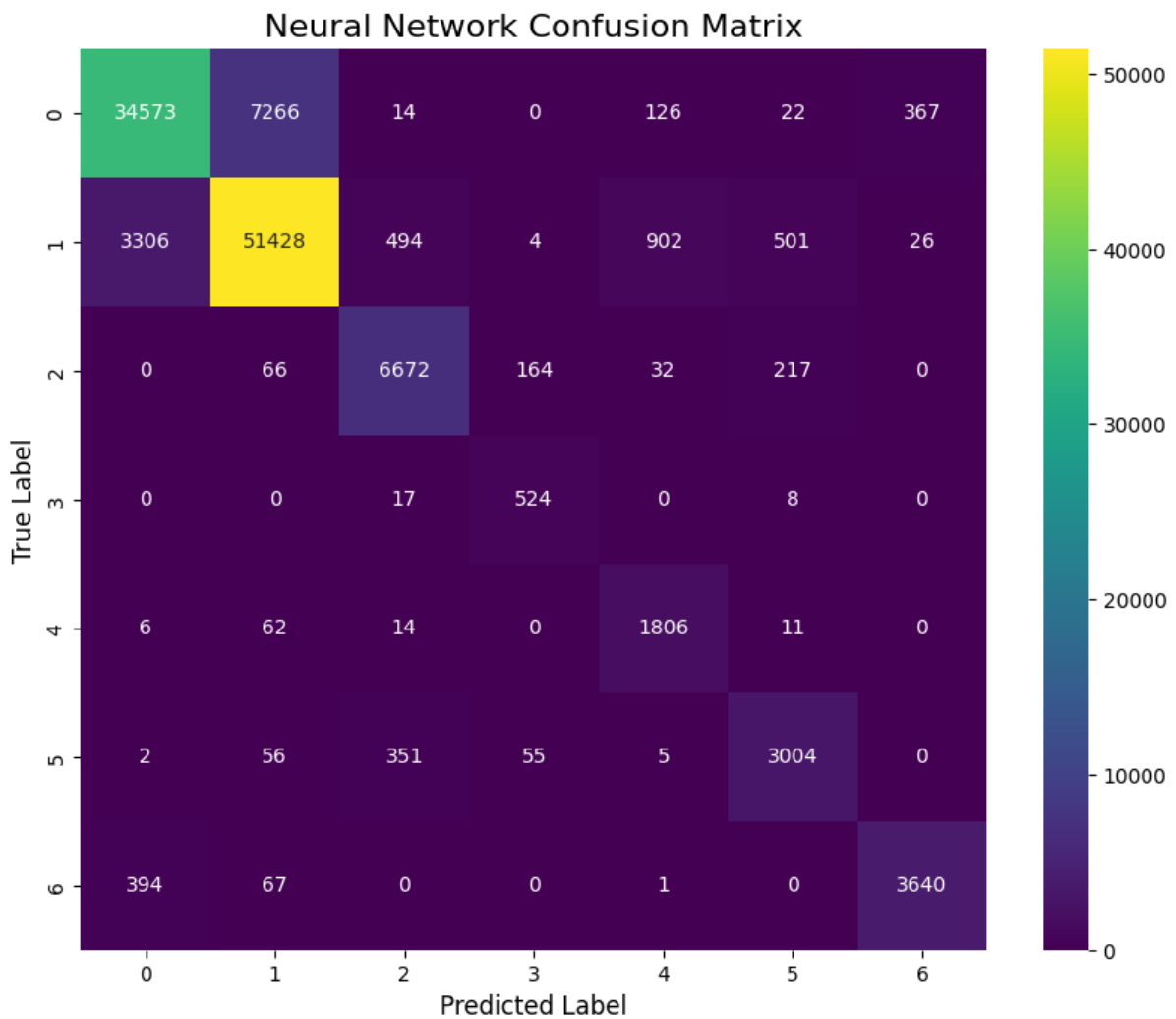


Figure 12: Neural Network confusion matrix

3.2.4 Impact of Class Imbalance and SMOTE

The original dataset suffered from severe class imbalance. Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data to mitigate this issue.

- **Clustering Methods:** SMOTE did not resolve the fundamental issue that the data’s inherent cluster structure does not map cleanly to the Cover_Type labels. The synthetic data failed to provide meaningful structure that would lead to correct classification.
- **Logistic Regression:** SMOTE was partially effective, evidenced by high recall for minority classes. However, the resulting low precision suggests that the linear model was unable to effectively separate the densely packed, synthetically balanced classes, leading to numerous false positives.
- **Neural Network:** SMOTE was **highly effective**. By providing a balanced representation of all classes, the NN was able to leverage its non-linear capacity to learn robust, generalized patterns for every Cover_Type. This resulted in high and balanced precision and recall across the board.

3.3 Conclusion of Forest cover type

The comparative analysis conclusively establishes the **Neural Network** as the superior model for the Forest Cover Type prediction, achieving an overall accuracy of 87.47% and providing robust, balanced performance across all seven classes. The NN’s success was heavily reliant on its non-linear modeling capacity combined with the use of SMOTE to address the severe class imbalance in the training data. The analysis also serves as an important reminder that unsupervised methods, while useful for data exploration, are not reliable substitutes for supervised classifiers when class labels do not directly align with underlying data clusters.

4 Conclusion

This project successfully implemented, tuned, and evaluated a range of classification models—Logistic Regression, SVM, and Neural Networks—across two fundamentally different datasets, yielding distinct optimal solutions for each challenge.

4.1 Comparative Model Efficacy Across Datasets

- **Binary Classification (Smoker Status):** The task, characterized by numerous engineered features derived from biosignals, was best solved by a **Tuned Logistic Regression** model, which achieved the highest accuracy of 0.7516 (Table 1). This demonstrates that for datasets where the relationship between features and the binary target can be captured effectively through a sophisticated linear/generalized linear boundary, linear models remain efficient and competitive with non-linear models like SVM and NN. The Neural Network struggled with overfitting despite feature scaling, suggesting that its complexity was not immediately beneficial without extensive regularization.
- **Multiclass Classification (Forest Cover Type):** The high-dimensional, multi-class nature of this problem, complicated by severe class imbalance, demanded a model capable of learning highly complex, non-linear decision boundaries. The **Neural Network** was the undisputed champion, achieving an accuracy of 0.8747 with robust performance across all seven cover types. This outcome underscores the superiority of deep learning architectures in modeling intricate feature interactions and overcoming the limitations of linear and kernel-based methods in complex, large-scale classification tasks.

The project successfully achieved its goal, concluding that model complexity should be tailored to data complexity, with linear models proving sufficient for the refined Smoker Status data, and the Neural Network being necessary for the highly complex, multi-class Forest Cover Type challenge.